2.36 Integration by Parts

Worked Solutions

1a)

$$\int xe^x\,dx$$

$$\begin{aligned} u &= x & \frac{dv}{dx} &= e^x \\ \frac{du}{dx} &= 1 & v &= e^x \end{aligned}$$

$$\begin{aligned} \int xe^x\,dx &= xe^x - \int e^x \times 1\,dx = xe^x - \int e^x\,dx \\ &= xe^x - e^x + c \end{aligned}$$

1b)

$$\int (x-1)e^x\,dx$$

$$\begin{aligned} u &= x-1 & \frac{dv}{dx} &= e^x \\ \frac{du}{dx} &= 1 & v &= e^x \end{aligned}$$

$$\begin{aligned} \int (x-1)e^x\,dx &= (x-1)e^x - \int e^x \times 1\,dx \\ &= (x-1)e^x - \int e^x\,dx \\ &= (x-1)e^x - e^x + c = xe^x - 2e^x + c \end{aligned}$$

1c)

$$\int (2x+5)e^x\,dx$$

$$\begin{aligned} u &= 2x+5 & \frac{dv}{dx} &= e^x \\ \frac{du}{dx} &= 2 & v &= e^x \end{aligned}$$

$$\begin{aligned} \int (2x+5)e^x\,dx &= (2x+5)e^x - \int e^x \times 2\,dx \\ &= (2x+5)e^x - \int 2e^x\,dx \\ &= (2x+5)e^x - 2e^x + c = 2xe^x + 3e^x + c \end{aligned}$$

2)

$$\int_{-\frac{1}{3}}^{\frac{1}{3}} xe^{3x}\,dx$$

$$\begin{aligned} u &= x & \frac{dv}{dx} &= e^{3x} \\ \frac{du}{dx} &= 1 & v &= \frac{1}{3}e^{3x} \end{aligned}$$

$$\begin{aligned}
\int_{-\frac{1}{3}}^{\frac{1}{3}} xe^{3x}\,dx &= \left[\tfrac{1}{3}xe^{3x}\right]_{-\frac{1}{3}}^{\frac{1}{3}} - \int_{-\frac{1}{3}}^{\frac{1}{3}} \tfrac{1}{3}e^{3x} \times 1\,dx \\
&= \left[\tfrac{1}{3}xe^{3x}\right]_{-\frac{1}{3}}^{\frac{1}{3}} - \int_{-\frac{1}{3}}^{\frac{1}{3}} \tfrac{1}{3}e^{3x}\,dx \\
&= \left[\tfrac{1}{3}xe^{3x}\right]_{-\frac{1}{3}}^{\frac{1}{3}} - \left[\tfrac{1}{9}e^{3x}\right]_{-\frac{1}{3}}^{\frac{1}{3}} = \left[\tfrac{1}{3}xe^{3x} - \tfrac{1}{9}e^{3x}\right]_{-\frac{1}{3}}^{\frac{1}{3}} \\
&= \left(\tfrac{1}{3} \times \tfrac{1}{3}e^{3\times\frac{1}{3}} - \tfrac{1}{9}e^{3\times\frac{1}{3}}\right) \\
&\quad - \left(\tfrac{1}{3} \times (-\tfrac{1}{3})e^{3\times(-\frac{1}{3})} - \tfrac{1}{9}e^{3\times(-\frac{1}{3})}\right) \\
&= \left(\tfrac{1}{9}e^{1} - \tfrac{1}{9}e^{1}\right) - \left(-\tfrac{1}{9}e^{-1} - \tfrac{1}{9}e^{-1}\right) \\
&= (0) - \left(-\tfrac{2}{9}e^{-1}\right) = \tfrac{2}{9}e^{-1} = \frac{2}{9e}
\end{aligned}$$

3)

$$\int_{-1}^{0} \frac{x+1}{3e^{x}}\,dx = \int_{-1}^{0} \tfrac{1}{3}(x+1)e^{-x}\,dx$$

$$\begin{aligned} u &= \frac{1}{3}(x+1) & \frac{dv}{dx} &= e^{-x} \\ \frac{du}{dx} &= \frac{1}{3} & v &= -e^{-x} \end{aligned}$$

$$\begin{aligned}
\int_{-1}^{0} \frac{x+1}{3e^{x}}\,dx &= \left[-\tfrac{1}{3}(x+1)e^{-x}\right]_{-1}^{0} - \int_{-1}^{0} -e^{-x} \times \tfrac{1}{3}\,dx \\
&= \left[-\tfrac{1}{3}(x+1)e^{-x}\right]_{-1}^{0} + \int_{-1}^{0} \tfrac{1}{3}e^{-x}\,dx \\
&= \left[-\tfrac{1}{3}(x+1)e^{-x}\right]_{-1}^{0} + \left[-\tfrac{1}{3}e^{-x}\right]_{-1}^{0} \\
&= \left[-\tfrac{1}{3}(x+1)e^{-x} - \tfrac{1}{3}e^{-x}\right]_{-1}^{0} \\
&= \left(-\tfrac{1}{3}(0+1)e^{0} - \tfrac{1}{3}e^{0}\right) \\
&\quad - \left(-\tfrac{1}{3}(-1+1)e^{-(-1)} - \tfrac{1}{3}e^{-(-1)}\right) \\
&= \left(-\tfrac{1}{3} - \tfrac{1}{3}\right) - \left(0 - \tfrac{1}{3}e^{1}\right) = \tfrac{1}{3}e - \tfrac{2}{3}
\end{aligned}$$

4)

$$\frac{dy}{dx} = 2x\sqrt{e^{x}} \Rightarrow \frac{dy}{dx} = 2xe^{\frac{1}{2}x}$$

$$u = 2x \quad \frac{dv}{dx} = e^{\frac{1}{2}x}$$
$$\frac{du}{dx} = 2 \quad v = 2e^{\frac{1}{2}x}$$

$$y = 4xe^{\frac{1}{2}x} - \int 2e^{\frac{1}{2}x} \times 2\,dx \Rightarrow y = 4xe^{\frac{1}{2}x} - \int 4e^{\frac{1}{2}x}\,dx$$
$$\Rightarrow y = 4xe^{\frac{1}{2}x} - 8e^{\frac{1}{2}x} + c$$

When $x = \ln 4$, $y = 2$:

$$2 = 4 \times \ln 4 \times e^{\frac{1}{2}\ln 4} - 8e^{\frac{1}{2}\ln 4} + c$$
$$\Rightarrow 2 = 4\ln 4 \times e^{\ln 2} - 8 \times e^{\ln 2} + c$$
$$\Rightarrow 2 = 8\ln 4 - 8 \times 2 + c$$
$$\Rightarrow c = 18 - 8\ln 4$$
$$y = 4xe^{\frac{1}{2}x} - 8e^{\frac{1}{2}x} - 8\ln 4 + 18$$
$$\therefore y = 4(x-2)\sqrt{e^x} + 8\ln 4 + 18$$

5a)

$$\int x \sin x\,dx$$

$$u = x \quad \frac{dv}{dx} = \sin x$$
$$\frac{du}{dx} = 1 \quad v = -\cos x$$

$$\int x \sin x\,dx = x \times -\cos x - \int -\cos x \times 1\,dx$$
$$= -x\cos x + \int \cos x\,dx = -x\cos x + \sin x + c$$

5b)

$$\int (x + \tfrac{\pi}{3}) \sin x\,dx$$

$$u = x + \frac{\pi}{3} \quad \frac{dv}{dx} = \sin x$$
$$\frac{du}{dx} = 1 \quad v = -\cos x$$

$$\int (x + \tfrac{\pi}{3}) \sin x\,dx = (x + \tfrac{\pi}{3}) \times -\cos x - \int -\cos x \times 1\,dx$$
$$= -(x + \tfrac{\pi}{3})\cos x + \int \cos x\,dx$$
$$= -(x + \tfrac{\pi}{3})\cos x + \sin x + c$$

6)

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x \sin 2x \, dx$$

$$\begin{aligned} u &= x & \frac{dv}{dx} &= \sin 2x \\ \frac{du}{dx} &= 1 & v &= -\frac{1}{2}\cos 2x \end{aligned}$$

$$\begin{aligned} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x \sin 2x \, dx &= \left[-\tfrac{1}{2}x\cos 2x\right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} - \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} -\tfrac{1}{2}\cos 2x \times 1 \, dx \\ &= \left[-\tfrac{1}{2}x\cos 2x\right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \tfrac{1}{2}\cos 2x \, dx \\ &= \left[-\tfrac{1}{2}x\cos 2x\right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} + \left[\tfrac{1}{4}\sin 2x\right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \\ &= \left[-\tfrac{1}{2}x\cos 2x + \tfrac{1}{4}\sin 2x\right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \\ &= \left(-\tfrac{1}{2} \times \tfrac{\pi}{2}\cos(2 \times \tfrac{\pi}{2}) + \tfrac{1}{4}\sin(2 \times \tfrac{\pi}{2})\right) \\ &\quad - \left(-\tfrac{1}{2} \times (-\tfrac{\pi}{2})\cos(2 \times (-\tfrac{\pi}{2})) + \tfrac{1}{4}\sin(2 \times (-\tfrac{\pi}{2}))\right) \\ &= (\tfrac{\pi}{4} + 0) - (-\tfrac{\pi}{4} + 0) = \tfrac{\pi}{2} \end{aligned}$$

7a)

$$\int x \cos x \, dx$$

$$\begin{aligned} u &= x & \frac{dv}{dx} &= \cos x \\ \frac{du}{dx} &= 1 & v &= \sin x \end{aligned}$$

$$\begin{aligned} \int x \cos x \, dx &= x \times \sin x - \int \sin x \times 1 \, dx \\ &= x \sin x - \int \sin x \, dx \\ &= x \sin x + \cos x + c \end{aligned}$$

7b)

$$\int x \cos x \, dx$$

$$\begin{aligned} u &= x - \frac{\pi}{6} & \frac{dv}{dx} &= \cos x \\ \frac{du}{dx} &= 1 & v &= \sin x \end{aligned}$$

$$\begin{aligned} \int (x - \tfrac{\pi}{6}) \cos x \, dx &= (x - \tfrac{\pi}{6}) \times \sin x - \int \sin x \times 1 \, dx \\ &= (x - \tfrac{\pi}{6}) \sin x - \int \sin x \, dx \\ &= (x - \tfrac{\pi}{6}) \sin x + \cos x + c \end{aligned}$$

8)

$$\int_{\frac{3\pi}{2}}^{\frac{5\pi}{2}} \left(x - \tfrac{2\pi}{3}\right) \cos x \, dx$$

$$\begin{aligned} u &= x - \frac{2\pi}{3} & \frac{dv}{dx} &= \cos x \\ \frac{du}{dx} &= 1 & v &= \sin x \end{aligned}$$

$$\begin{aligned} \int_{\frac{3\pi}{2}}^{\frac{5\pi}{2}} \left(x - \tfrac{2\pi}{3}\right) \cos x \, dx &= \left[\left(x - \tfrac{2\pi}{3}\right) \sin x\right]_{\frac{3\pi}{2}}^{\frac{5\pi}{2}} - \int_{\frac{3\pi}{2}}^{\frac{5\pi}{2}} \sin x \times 1 \, dx \\ &= \left[\left(x - \tfrac{2\pi}{3}\right) \sin x\right]_{\frac{3\pi}{2}}^{\frac{5\pi}{2}} - \int_{\frac{3\pi}{2}}^{\frac{5\pi}{2}} \sin x \, dx \\ &= \left[\left(x - \tfrac{2\pi}{3}\right) \sin x\right]_{\frac{3\pi}{2}}^{\frac{5\pi}{2}} - \left[-\cos x\right]_{\frac{3\pi}{2}}^{\frac{5\pi}{2}} \\ &= \left[\left(x - \tfrac{2\pi}{3}\right) \sin x + \cos x\right]_{\frac{3\pi}{2}}^{\frac{5\pi}{2}} \\ &= \left(\left(\tfrac{5\pi}{2} - \tfrac{2\pi}{3}\right) \sin \tfrac{5\pi}{2} + \cos \tfrac{5\pi}{2}\right) \\ &\quad - \left(\left(\tfrac{3\pi}{2} - \tfrac{2\pi}{3}\right) \sin \tfrac{3\pi}{2} + \cos \tfrac{3\pi}{2}\right) \\ &= \left(\tfrac{11\pi}{6} + 0\right) - \left(-\tfrac{5\pi}{6} + 0\right) = \tfrac{11\pi}{6} + \tfrac{5\pi}{6} = \tfrac{8\pi}{3} \end{aligned}$$

9)

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (2x - \tfrac{\pi}{4}) \sin 2x \, dx$$

$$\begin{aligned} u &= 2x - \frac{\pi}{4} & \frac{dv}{dx} &= \sin 2x \\ \frac{du}{dx} &= 2 & v &= -\frac{1}{2} \cos 2x \end{aligned}$$

$$\begin{aligned} \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (2x - \tfrac{\pi}{4}) \sin 2x \, dx &= \left[-\tfrac{1}{2} (2x - \tfrac{\pi}{4}) \cos 2x\right]_{\frac{\pi}{4}}^{\frac{\pi}{2}} - \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} -\tfrac{1}{2} \cos 2x \times 2 \, dx \\ &= \left[-\tfrac{1}{2} (2x - \tfrac{\pi}{4}) \cos 2x\right]_{\frac{\pi}{4}}^{\frac{\pi}{2}} + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos 2x \, dx \\ &= \left[-\tfrac{1}{2} (2x - \tfrac{\pi}{4}) \cos 2x + \tfrac{1}{2} \sin 2x\right]_{\frac{\pi}{4}}^{\frac{\pi}{2}} \\ &= \left(-\tfrac{1}{2} (2 \times \tfrac{\pi}{2} - \tfrac{\pi}{4}) \cos(2 \times \tfrac{\pi}{2}) + \tfrac{1}{2} \sin(2 \times \tfrac{\pi}{2})\right) \\ &\quad - \left(-\tfrac{1}{2} (2 \times \tfrac{\pi}{4} - \tfrac{\pi}{4}) \cos(2 \times \tfrac{\pi}{4}) + \tfrac{1}{2} \sin(2 \times \tfrac{\pi}{4})\right) \\ &= \left(\tfrac{3\pi}{8} + 0\right) - \left(0 + \tfrac{1}{2}\right) = \tfrac{3\pi}{8} - \tfrac{1}{2} \end{aligned}$$

10)

$$\int_{-\frac{\pi}{8}}^{\frac{\pi}{8}} (x + \tfrac{\pi}{8}) \cos 4x \, dx$$

$$u = x + \frac{\pi}{8} \quad \frac{dv}{dx} = \cos 4x$$
$$\frac{du}{dx} = 1 \qquad v = \frac{1}{4}\sin 4x$$

$$\begin{aligned}
\int_{-\frac{\pi}{8}}^{\frac{\pi}{8}} (x+\tfrac{\pi}{8})\cos 4x\,dx &= \left[\tfrac{1}{4}(x+\tfrac{\pi}{8})\sin 4x\right]_{-\frac{\pi}{8}}^{\frac{\pi}{8}} - \int_{-\frac{\pi}{8}}^{\frac{\pi}{8}} \tfrac{1}{4}\sin 4x\,dx \\
&= \left[\tfrac{1}{4}(x+\tfrac{\pi}{8})\sin 4x\right]_{-\frac{\pi}{8}}^{\frac{\pi}{8}} + \left[\tfrac{1}{16}\cos 4x\right]_{-\frac{\pi}{8}}^{\frac{\pi}{8}} \\
&= \left[\tfrac{1}{4}(x+\tfrac{\pi}{8})\sin 4x + \tfrac{1}{16}\cos 4x\right]_{-\frac{\pi}{8}}^{\frac{\pi}{8}} \\
&= \left(\tfrac{1}{4}(\tfrac{\pi}{8}+\tfrac{\pi}{8})\sin(4\times\tfrac{\pi}{8}) + \tfrac{1}{16}\cos(4\times\tfrac{\pi}{8})\right) \\
&\quad - \left(\tfrac{1}{4}(-\tfrac{\pi}{8}+\tfrac{\pi}{8})\sin(4\times-\tfrac{\pi}{8}) + \tfrac{1}{16}\cos(4\times-\tfrac{\pi}{8})\right) \\
&= (\tfrac{\pi}{16}+0) - (0+0) = \tfrac{\pi}{16}
\end{aligned}$$

11)

$$\frac{dy}{dx} = 64x\sin 8x$$
$$u = 64x \quad \frac{dv}{dx} = \sin 8x$$
$$\frac{du}{dx} = 64 \quad v = -\frac{1}{8}\cos 8x$$

$$y = 64x \times -\tfrac{1}{8}\cos 8x - \int -\tfrac{1}{8}\cos 8x \times 64\,dx$$
$$\Rightarrow y = -8x\cos 8x + \int 8\cos 8x\,dx$$
$$\Rightarrow y = -8x\cos 8x + \sin 8x + c$$

When $x = \frac{\pi}{16}$, $y = 1$:

$$1 = -8 \times \tfrac{\pi}{16} \times \cos(8 \times \tfrac{\pi}{16}) + \sin(8 \times \tfrac{\pi}{16}) + c$$
$$\Rightarrow 1 = 0 + 1 + c \Rightarrow c = 0$$
$$\therefore y = \sin 8x - 8x\cos 8x$$

12)

$$\frac{dy}{dx} = 6x\cos\tfrac{1}{3}x$$
$$u = 6x \quad \frac{dv}{dx} = \cos\frac{1}{3}x$$
$$\frac{du}{dx} = 6 \quad v = 3\sin\frac{1}{3}x$$

$$y = 6x \times 3\sin\tfrac{1}{3}x - \int 3\sin\tfrac{1}{3}x \times 6\,dx$$
$$\Rightarrow y = 18x\sin\tfrac{1}{3}x - \int 18\sin\tfrac{1}{3}x\,dx$$
$$\Rightarrow y = 18x\sin\tfrac{1}{3}x + 54\cos\tfrac{1}{3}x + c$$

When $x = \pi$, $y = -1$:

$$-1 = 18\pi \sin(\tfrac{1}{3} \times \pi) + 54\cos(\tfrac{1}{3} \times \pi) + c$$
$$\Rightarrow -1 = 9\sqrt{3}\pi + 27 + c \Rightarrow c = -9\sqrt{3}\pi - 28$$
$$\therefore y = 18x \sin \tfrac{1}{3}x + 54 \cos \tfrac{1}{3}x - 9\sqrt{3}\pi - 28$$

13a)

$$\int x \ln x \, dx$$

$$u = \ln x \quad \frac{dv}{dx} = x$$
$$\frac{du}{dx} = \frac{1}{x} \quad v = \frac{1}{2}x^2$$

$$\int x \ln x \, dx = \ln x \times \tfrac{1}{2}x^2 - \int \tfrac{1}{2}x^2 \times \tfrac{1}{x} \, dx$$
$$= \tfrac{1}{2}x^2 \ln x - \int \tfrac{1}{2}x \, dx$$
$$= \tfrac{1}{2}x^2 \ln x - \tfrac{1}{4}x^2 + c$$

13b)

$$\int x^2 \ln x \, dx$$

$$u = \ln x \quad \frac{dv}{dx} = x^2$$
$$\frac{du}{dx} = \frac{1}{x} \quad v = \frac{1}{3}x^3$$

$$\int x^2 \ln x \, dx = \ln x \times \tfrac{1}{3}x^3 - \int \tfrac{1}{3}x^3 \times \tfrac{1}{x} \, dx$$
$$= \tfrac{1}{3}x^3 \ln x - \int \tfrac{1}{3}x^2 \, dx$$
$$= \tfrac{1}{3}x^3 \ln x - \tfrac{1}{9}x^3 + c$$

13c)

$$\int (x+4) \ln x \, dx$$

$$u = \ln x \quad \frac{dv}{dx} = x + 4$$
$$\frac{du}{dx} = \frac{1}{x} \quad v = \frac{1}{2}x^2 + 4x$$

$$\int (x+4) \ln x \, dx = \left(\tfrac{1}{2}x^2 + 4x\right) \times \ln x - \int \left(\tfrac{1}{2}x^2 + 4x\right) \times \tfrac{1}{x} \, dx$$
$$= \left(\tfrac{1}{2}x^2 + 4x\right) \ln x - \int \left(\tfrac{1}{2}x + 4\right) \, dx$$
$$= \left(\tfrac{1}{2}x^2 + 4x\right) \ln x - \left(\tfrac{1}{4}x^2 + 4x\right) + c$$

13d)

$$\int (x^3 + 3x) \ln x \, dx$$

$$\begin{aligned} u &= \ln x & \frac{dv}{dx} &= x^3 + 3x \\ \frac{du}{dx} &= \frac{1}{x} & v &= \frac{1}{4}x^4 + \frac{3}{2}x^2 \end{aligned}$$

$$\begin{aligned} \int (x^2+1) \ln x \, dx &= \ln x \times \left(\tfrac{1}{4}x^4 + \tfrac{3}{2}x^2\right) - \int \left(\tfrac{1}{4}x^4 + \tfrac{3}{2}x^2\right) \times \tfrac{1}{x} \, dx \\ &= \left(\tfrac{1}{4}x^4 + \tfrac{3}{2}x^2\right) \ln x - \int \left(\tfrac{1}{4}x^3 + \tfrac{3}{2}x\right) \, dx \\ &= \tfrac{1}{4}x^2(x^2+6) \ln x - \left(\tfrac{1}{16}x^4 + \tfrac{3}{4}x^2\right) + c \end{aligned}$$

14)

$$\int_1^2 \frac{\ln 8x}{x^4} \, dx = \int_1^2 x^{-4} \ln 8x \, dx$$

$$\begin{aligned} u &= \ln 8x & \frac{dv}{dx} &= x^{-4} \\ \frac{du}{dx} &= \frac{1}{x} & v &= -\frac{1}{3}x^{-3} \end{aligned}$$

$$\begin{aligned} \int_1^2 \frac{\ln 8x}{x^4} \, dx &= \left[\ln 8x \times -\tfrac{1}{3}x^{-3}\right]_1^2 - \int_1^2 -\tfrac{1}{3}x^{-3} \times \tfrac{1}{x} \, dx \\ &= \left[-\tfrac{1}{3}x^{-3} \ln 8x\right]_1^2 + \int_1^2 \tfrac{1}{3}x^{-4} \, dx \\ &= \left[-\tfrac{1}{3}x^{-3} \ln 8x\right]_1^2 + \left[-\tfrac{1}{9}x^{-3}\right]_1^2 \\ &= \left[-\tfrac{1}{3}x^{-3} \ln 8x - \tfrac{1}{9}x^{-3}\right]_1^2 \\ &= \left(-\tfrac{1}{3} \times 2^{-3} \ln(8 \times 2) - \tfrac{1}{9} \times 2^{-3}\right) \\ &\quad - \left(-\tfrac{1}{3} \times 1^{-3} \ln(8 \times 1) - \tfrac{1}{9} \times 1^{-3}\right) \\ &= \left(-\tfrac{1}{24} \ln 16 - \tfrac{1}{72}\right) - \left(-\tfrac{1}{3} \ln 8 - \tfrac{1}{9}\right) \\ &= \tfrac{1}{3} \ln 8 - \tfrac{1}{24} \ln 16 + \tfrac{7}{72} \end{aligned}$$

15)

$$\frac{dy}{dx} = 4x \ln \tfrac{1}{2}x$$

$$\begin{aligned} u &= \ln \frac{1}{2}x & \frac{dv}{dx} &= 4x \\ \frac{du}{dx} &= \frac{1}{x} & v &= 2x^2 \end{aligned}$$

$$y = \ln \tfrac{1}{2}x \times 2x^2 - \int 2x^2 \times \tfrac{1}{x} \, dx$$

$$\Rightarrow y = 2x^2 \ln \tfrac{1}{2}x - \int 2x \, dx$$

$$\Rightarrow y = 2x^2 \ln \tfrac{1}{2}x - x^2 + c$$

When $x = 2e$, $y = 1$:

$$\begin{gathered}
1 = 2(2e)^2 \ln(\tfrac{1}{2} \times 2e) - (2e)^2 + c \\
\Rightarrow 1 = 8e^2 - 4e^2 + c \Rightarrow c = 1 - 4e^2 \\
\therefore y = 2x^2 \ln \tfrac{1}{2}x - x^2 + 1 - 4e^2 \\
\Rightarrow y = -x^2 + 2x^2 \ln \tfrac{1}{2}x - 4e^2 + 1
\end{gathered}$$

16a)

$$\int x2^x \, dx$$

$$\begin{aligned}
u &= x & \frac{dv}{dx} &= 2^x \\
\frac{du}{dx} &= 1 & v &= \frac{1}{\ln 2}2^x
\end{aligned}$$

$$\begin{aligned}
\int x2^x \, dx &= x \times \tfrac{1}{\ln 2}2^x - \int \tfrac{1}{\ln 2}2^x \times 1 \, dx \\
&= \tfrac{1}{\ln 2}x2^x - \int \tfrac{1}{\ln 2}2^x \, dx \\
&= \tfrac{1}{\ln 2}x2^x - \tfrac{1}{(\ln 2)^2}2^x + c
\end{aligned}$$

16b)

$$\int (3x + 9)3^x \, dx$$

$$\begin{aligned}
u &= 3x + 9 & \frac{dv}{dx} &= 3^x \\
\frac{du}{dx} &= 3 & v &= \frac{1}{\ln 3}3^x
\end{aligned}$$

$$\begin{aligned}
\int (3x + 9)3^x \, dx &= (3x + 9) \times \tfrac{1}{\ln 3}3^x - \int \tfrac{1}{\ln 3}3^x \times 3 \, dx \\
&= \tfrac{1}{\ln 3}(3x + 9)3^x - \int \tfrac{3}{\ln 3}3^x \, dx \\
&= \tfrac{1}{\ln 3}(3x + 9)3^x - \tfrac{3}{(\ln 3)^2}3^x + c
\end{aligned}$$

17)

$$\int_0^1 (x + 2)2^x \, dx$$

$$\begin{aligned}
u &= x + 2 & \frac{dv}{dx} &= 2^x \\
\frac{du}{dx} &= 1 & v &= \frac{1}{\ln 2}2^x
\end{aligned}$$

$$\int_0^1 (x + 2)2^x \, dx = \left[(x + 2) \times \tfrac{1}{\ln 2}2^x\right]_0^1 - \int_0^1 \tfrac{1}{\ln 2}2^x \times 1 \, dx$$

$$
\begin{aligned}
&= \left[\tfrac{1}{\ln 2}(x+2)2^x\right]_0^1 - \int_0^1 \tfrac{1}{\ln 2}2^x\,dx \\
&= \left[\tfrac{1}{\ln 2}(x+2)2^x\right]_0^1 - \left[\tfrac{1}{(\ln 2)^2}2^x\right]_0^1 \\
&= \left[\tfrac{1}{\ln 2}(x+2)2^x - \tfrac{1}{(\ln 2)^2}2^x\right]_0^1 \\
&= \left(\tfrac{1}{\ln 2}(1+2)2^1 - \tfrac{1}{(\ln 2)^2}2^1\right) \\
&\quad - \left(\tfrac{1}{\ln 2}(0+2)2^0 - \tfrac{1}{(\ln 2)^2}2^0\right) \\
&= \left(\tfrac{6}{\ln 2} - \tfrac{2}{(\ln 2)^2}\right) - \left(\tfrac{2}{\ln 2} - \tfrac{1}{(\ln 2)^2}\right) \\
&= \tfrac{4}{\ln 2} - \tfrac{1}{(\ln 2)^2} = \frac{4\ln 2 - 1}{(\ln 2)^2}
\end{aligned}
$$

18)

$$
\int_1^2 10x5^{x-1}\,dx = \int_1^2 10x5^x \times 5^{-1}\,dx = \int_1^2 2x5^x\,dx
$$

$$
\begin{aligned}
u &= 2x & \frac{dv}{dx} &= 5^x \\
\frac{du}{dx} &= 2 & v &= \frac{1}{\ln 5}5^x
\end{aligned}
$$

$$
\begin{aligned}
\int_1^2 2x5^x\,dx &= \left[2x \times \tfrac{1}{\ln 5}5^x\right]_1^2 - \int_1^2 \tfrac{1}{\ln 5}5^x \times 2\,dx \\
&= \left[\tfrac{2}{\ln 5}x5^x\right]_1^2 - \int_1^2 \tfrac{2}{\ln 5}5^x\,dx \\
&= \left[\tfrac{2}{\ln 5}x5^x\right]_1^2 - \left[\tfrac{2}{(\ln 5)^2}5^x\right]_1^2 \\
&= \left[\tfrac{2}{\ln 5}x5^x - \tfrac{2}{(\ln 5)^2}5^x\right]_1^2 \\
&= \left(\tfrac{2}{\ln 5} \times 2 \times 5^2 - \tfrac{2}{(\ln 5)^2}5^2\right) \\
&\quad - \left(\tfrac{2}{\ln 5} \times 1 \times 5^1 - \tfrac{2}{(\ln 5)^2}5^1\right) \\
&= \left(\tfrac{100}{\ln 5} - \tfrac{50}{(\ln 5)^2}\right) - \left(\tfrac{10}{\ln 5} - \tfrac{10}{(\ln 5)^2}\right) \\
&= \tfrac{90}{\ln 5} - \tfrac{40}{(\ln 5)^2}
\end{aligned}
$$

19)

$$
\frac{dy}{dx} = \frac{x}{4^x} \Rightarrow \frac{dy}{dx} = x4^{-x}
$$

$$
\begin{aligned}
u &= x & \frac{dv}{dx} &= 4^{-x} \\
\frac{du}{dx} &= 1 & v &= -\frac{1}{\ln 4}4^{-x}
\end{aligned}
$$

$$
\begin{aligned}
y &= x \times -\tfrac{1}{\ln 4}4^{-x} - \int -\tfrac{1}{\ln 4}4^{-x} \times 1\,dx \\
\Rightarrow y &= -\tfrac{1}{\ln 4}x4^{-x} + \int \tfrac{1}{\ln 4}4^{-x}\,dx \\
\Rightarrow y &= -\tfrac{1}{\ln 4}x4^{-x} + \int \tfrac{1}{\ln 4}4^{-x}\,dx
\end{aligned}
$$

$$\Rightarrow y = -\tfrac{1}{\ln 4}x4^{-x} - \tfrac{1}{(\ln 4)^2}4^{-x} + c$$

When $x = \ln 4$, $y = 1$:

$$\begin{aligned}
\Rightarrow 1 &= -\tfrac{1}{\ln 4}(\ln 4)4^{-\ln 4} - \tfrac{1}{(\ln 4)^2}4^{-\ln 4} + c \\
1 &= -4^{-\ln 4} - \tfrac{1}{(\ln 4)^2}4^{-\ln 4} + c \\
\Rightarrow c &= 1 + 4^{-\ln 4} + \tfrac{1}{(\ln 4)^2}4^{-\ln 4} \\
\Rightarrow c &= 1 + 4^{-\ln 4}\left(1 + \tfrac{1}{(\ln 4)^2}\right) \\
\therefore y &= -\tfrac{1}{\ln 4}x4^{-x} - \tfrac{1}{(\ln 4)^2}4^{-x} + 1 + 4^{-\ln 4}\left(1 + \tfrac{1}{(\ln 4)^2}\right)
\end{aligned}$$

20)

$$\int_0^{\frac{\pi}{3}} x\sec^2 x\,dx$$

$$\begin{aligned}
u = x \quad & \frac{dv}{dx} = \sec^2 x \\
\frac{du}{dx} = 1 \quad & v = \tan x
\end{aligned}$$

$$\begin{aligned}
\int_0^{\frac{\pi}{3}} x\sec^2 x\,dx &= [x \times \tan x]_0^{\frac{\pi}{3}} - \int_0^{\frac{\pi}{3}} \tan x \times 1\,dx \\
&= [x\tan x]_0^{\frac{\pi}{3}} - \int_0^{\frac{\pi}{3}} \tan x\,dx \\
&= [x\tan x]_0^{\frac{\pi}{3}} - [-\ln|\cos x|]_0^{\frac{\pi}{3}} \\
&= [x\tan x + \ln|\cos x|]_0^{\frac{\pi}{3}} \\
&= \left(\tfrac{\pi}{3} \times \tan\tfrac{\pi}{3} + \ln|\cos\tfrac{\pi}{3}|\right) - (0 \times \tan 0 + \ln|\cos 0|) \\
&= \left(\tfrac{\pi\sqrt{3}}{3} + \ln\tfrac{1}{2}\right) - (0 + \ln 1) \\
&= \tfrac{\pi\sqrt{3}}{3} + \ln\tfrac{1}{2} = \tfrac{\pi\sqrt{3}}{3} \times \tfrac{\sqrt{3}}{\sqrt{3}} + \ln 2^{-1} = \tfrac{\pi}{\sqrt{3}} - \ln 2
\end{aligned}$$

21)

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} 18x\,\mathrm{cosec}^2 x\,dx$$

$$\begin{aligned}
u = 18x \quad & \frac{dv}{dx} = \mathrm{cosec}^2 x \\
\frac{du}{dx} = 18 \quad & v = -\cot x
\end{aligned}$$

$$\begin{aligned}
\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} 18x\,\mathrm{cosec}^2 x\,dx &= [18x \times -\cot x]_{\frac{\pi}{6}}^{\frac{\pi}{3}} - \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} -\cot x \times 18\,dx \\
&= [-18x\cot x]_{\frac{\pi}{6}}^{\frac{\pi}{3}} + \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} 18\cot x\,dx \\
&= [-18x\cot x]_{\frac{\pi}{6}}^{\frac{\pi}{3}} + [18\ln|\sin x|]_{\frac{\pi}{6}}^{\frac{\pi}{3}} \\
&= [-18x\cot x + 18\ln|\sin x|]_{\frac{\pi}{6}}^{\frac{\pi}{3}}
\end{aligned}$$

$$
\begin{aligned}
&= \left(-18 \times \tfrac{\pi}{3} \times \cot \tfrac{\pi}{3} + 18 \ln |\sin \tfrac{\pi}{3}|\right) \\
&\quad - \left(-18 \times \tfrac{\pi}{6} \times \cot \tfrac{\pi}{6} + 18 \ln |\sin \tfrac{\pi}{6}|\right) \\
&= \left(-6\pi \times \tfrac{\sqrt{3}}{3} + 18 \ln \tfrac{\sqrt{3}}{2}\right) - \left(-3\pi \times \sqrt{3} + 18 \ln \tfrac{1}{2}\right) \\
&= \left(-2\pi\sqrt{3} + 18 \ln \tfrac{\sqrt{3}}{2}\right) - \left(-3\pi\sqrt{3} + 18 \ln \tfrac{1}{2}\right) \\
&= \pi\sqrt{3} + 18 \left(\ln \tfrac{\sqrt{3}}{2} - \ln \tfrac{1}{2}\right) \\
&= \pi\sqrt{3} + 18 \ln \sqrt{3}
\end{aligned}
$$

22a)

$$\int \ln x \, dx = \int \ln x \times 1 \, dx$$

$$
\begin{aligned}
u &= \ln x & \frac{dv}{dx} &= 1 \\
\frac{du}{dx} &= \frac{1}{x} & v &= x
\end{aligned}
$$

$$
\begin{aligned}
\int \ln x \, dx &= \ln x \times x - \int x \times \tfrac{1}{x} \, dx \\
&= x \ln x - \int 1 \, dx \\
&= x \ln x - x + c
\end{aligned}
$$

22b)

$$\int (\ln x)^2 \, dx = \int \ln x \times \ln x \, dx$$

$$
\begin{aligned}
u &= \ln x & \frac{dv}{dx} &= \ln x \\
\frac{du}{dx} &= \frac{1}{x} & v &= x \ln x - x
\end{aligned}
$$

$$
\begin{aligned}
\int (\ln x)^2 \, dx &= \ln x \times (x \ln x - x) - \int (x \ln x - x) \times \tfrac{1}{x} \, dx \\
&= x(\ln x)^2 - x \ln x - \int (\ln x - 1) \, dx \\
&= x(\ln x)^2 - x \ln x - (x \ln x - x - x) + c \\
&= x(\ln x)^2 - x \ln x - x \ln x + 2x + c \\
&= x(\ln x)^2 - 2x \ln x + 2x + c
\end{aligned}
$$

23)

$$\int_3^{e^2+2} \ln(x-2) \, dx$$

$$
\begin{aligned}
u &= \ln(x-2) & \frac{dv}{dx} &= 1 \\
\frac{du}{dx} &= \frac{1}{x-2} & v &= x
\end{aligned}
$$

$$\int_3^{e^2+2} \ln(x-2)\,dx = [\ln(x-2)\times x]_3^{e^2+2} - \int_3^{e^2+2} x \times \tfrac{1}{x-2}\,dx$$
$$= [x\ln(x-2)]_3^{e^2+2} - \int_3^{e^2+2} \tfrac{x}{x-2}\,dx$$

Focus on $\displaystyle\int_3^{e^2+2} \frac{x}{x-2}\,dx$ and use Integration by Substitution

$$u = x-2 \Rightarrow \frac{du}{dx} = 1 \Rightarrow dx = du$$
$$u = x-2 \Rightarrow x = u+2$$

Change limits:

$$x = e^2+2 \Rightarrow u = e^2+2-2 = e^2$$
$$x = 3 \Rightarrow u = 3-2 = 1$$

$$\int_3^{e^2+2} \frac{x}{x-2}\,dx = \int_1^{e^2} \frac{u+2}{u}\,du$$
$$= \int_1^{e^2} \left(\tfrac{u}{u} + \tfrac{2}{u}\right)\,du = \int_1^{e^2} \left(1 + \tfrac{2}{u}\right)\,du$$
$$= [u + 2\ln|u|]_1^{e^2}$$
$$= (e^2 + 2\ln|e^2|) - (1 + 2\ln|1|)$$
$$= (e^2 + 2\ln e^2) - (1 + 2\ln 1)$$
$$= (e^2+4) - (1+0) = e^2+3$$

$$\therefore \int_3^{e^2+2} \ln(x-2)\,dx = [x\ln(x-2)]_3^{e^2+2} - (e^2+3)$$
$$= ((e^2+2)\ln(e^2+2-2)) - (3\ln(3-2)) - e^2 - 3$$
$$= ((e^2+2)\ln e^2) - (3\ln 1) - e^2 - 3$$
$$= 2(e^2+2) - 0 - e^2 - 3 = e^2+1$$

24a)

$$\int x^2 e^x\,dx$$

$$u = x^2 \qquad \frac{dv}{dx} = e^x$$
$$\frac{du}{dx} = 2x \quad v = e^x$$

$$\int x^2 e^x\,dx = x^2e^x - \int e^x \times 2x\,dx$$
$$= x^2e^x - \int 2xe^x\,dx$$

$$u = 2x \quad \frac{dv}{dx} = e^x$$
$$\frac{du}{dx} = 2 \quad v = e^x$$

$$\begin{aligned}\int x^2 e^x \, dx &= x^2 e^x - \int 2xe^x \, dx \\ &= x^2 e^x - \left(2xe^x - \int e^x \times 2 \, dx\right) \\ &= x^2 e^x - 2xe^x + \int 2e^x \, dx \\ &= x^2 e^x - 2xe^x + 2e^x + c\end{aligned}$$

24b)

$$\int \frac{x^2}{e^x} \, dx = \int x^2 e^{-x} \, dx$$

$$\begin{aligned} u &= x^2 & \frac{dv}{dx} &= e^{-x} \\ \frac{du}{dx} &= 2x & v &= -e^{-x}\end{aligned}$$

$$\begin{aligned}\int x^2 e^{-x} \, dx &= x^2 \times -e^{-x} - \int -e^{-x} \times 2x \, dx \\ &= -x^2 e^{-x} + \int 2xe^{-x} \, dx\end{aligned}$$

$$\begin{aligned} u &= 2x & \frac{dv}{dx} &= e^{-x} \\ \frac{du}{dx} &= 2 & v &= -e^{-x}\end{aligned}$$

$$\begin{aligned}\int x^2 e^x \, dx &= -x^2 e^{-x} + \int 2xe^{-x} \, dx \\ &= -x^2 e^{-x} + \left(2x \times -e^{-x} - \int -e^{-x} \times 2 \, dx\right) \\ &= -x^2 e^{-x} + \left(-2xe^{-x} + \int 2e^{-x} \, dx\right) \\ &= -x^2 e^{-x} - 2xe^{-x} + \int 2e^{-x} \, dx \\ &= -x^2 e^{-x} - 2xe^{-x} - 2e^{-x} + c\end{aligned}$$

24c)

$$\int x^2 \sin x \, dx$$

$$\begin{aligned} u &= x^2 & \frac{dv}{dx} &= \sin x \\ \frac{du}{dx} &= 2x & v &= -\cos x\end{aligned}$$

$$\begin{aligned}\int x^2 \sin x \, dx &= x^2 \times -\cos x - \int -\cos x \times 2x \, dx \\ &= -x^2 \cos x + \int 2x \cos x \, dx\end{aligned}$$

$$\begin{aligned} u &= 2x & \frac{dv}{dx} &= \cos x \\ \frac{du}{dx} &= 2 & v &= \sin x\end{aligned}$$

$$\begin{aligned}\int x^2 \sin x \, dx &= -x^2 \cos x + \int 2x \cos x \, dx \\ &= -x^2 \cos x + \left(2x \sin x - \int \sin x \times 2 \, dx\right) \\ &= -x^2 \cos x + 2x \sin x - \int 2 \sin x \, dx \\ &= -x^2 \cos x + 2x \sin x + 2 \cos x + c\end{aligned}$$

24d)

$$\int x^2 2^x \, dx$$

$$\begin{aligned} u &= x^2 & \frac{dv}{dx} &= 2^x \\ \frac{du}{dx} &= 2x & v &= \frac{1}{\ln 2} 2^x \end{aligned}$$

$$\begin{aligned}\int x^2 2^x \, dx &= x^2 \times \tfrac{1}{\ln 2} 2^x - \int \tfrac{1}{\ln 2} 2^x \times 2x \, dx \\ &= \tfrac{1}{\ln 2} x^2 2^x - \int \tfrac{2}{\ln 2} x 2^x \, dx\end{aligned}$$

$$\begin{aligned} u &= \frac{2}{\ln 2} x & \frac{dv}{dx} &= 2^x \\ \frac{du}{dx} &= \frac{2}{\ln 2} & v &= \frac{1}{\ln 2} 2^x \end{aligned}$$

$$\begin{aligned}\int x^2 2^x \, dx &= \tfrac{1}{\ln 2} x^2 2^x - \int \tfrac{2}{\ln 2} x 2^x \, dx \\ &= \tfrac{1}{\ln 2} x^2 2^x - \left(\tfrac{2}{\ln 2} x \times \tfrac{1}{\ln 2} 2^x - \int \tfrac{1}{\ln 2} 2^x \times \tfrac{2}{\ln 2} \, dx\right) \\ &= \tfrac{1}{\ln 2} x^2 2^x - \left(\tfrac{2}{(\ln 2)^2} x 2^x - \int \tfrac{2}{(\ln 2)^2} 2^x \, dx\right) \\ &= \tfrac{1}{\ln 2} x^2 2^x - \tfrac{2}{(\ln 2)^2} x 2^x + \int \tfrac{2}{(\ln 2)^2} 2^x \, dx \\ &= \tfrac{1}{\ln 2} x^2 2^x - \tfrac{2}{(\ln 2)^2} x 2^x + \tfrac{2}{(\ln 2)^3} 2^x + c\end{aligned}$$

25)

$$\int_0^{\frac{1}{4}} 20x^2 e^{-4x} \, dx$$

$$\begin{aligned} u &= 20x^2 & \frac{dv}{dx} &= e^{-4x} \\ \frac{du}{dx} &= 40x & v &= -\frac{1}{4} e^{-4x} \end{aligned}$$

$$\begin{aligned}\int_0^{\frac{1}{4}} 20x^2 e^{-4x} \, dx &= \left[20x^2 \times -\tfrac{1}{4} e^{-4x}\right]_0^{\frac{1}{4}} - \int_0^{\frac{1}{4}} -\tfrac{1}{4} e^{-4x} \times 40x \, dx \\ &= \left[-5x^2 e^{-4x}\right]_0^{\frac{1}{4}} + \int_0^{\frac{1}{4}} 10x e^{-4x} \, dx\end{aligned}$$

$$\begin{aligned} u &= 10x & \frac{dv}{dx} &= e^{-4x} \\ \frac{du}{dx} &= 10 & v &= -\frac{1}{4}e^{-4x} \end{aligned}$$

$$\begin{aligned}
\int_0^{\frac{1}{4}} 20x^2e^{-4x}\,dx &= [-5x^2e^{-4x}]_0^{\frac{1}{4}} + \int_0^{\frac{1}{4}} 10xe^{-4x}\,dx \\
&= [-5x^2e^{-4x}]_0^{\frac{1}{4}} + \left(\left[10x \times -\tfrac{1}{4}e^{-4x}\right]_0^{\frac{1}{4}} - \int_0^{\frac{1}{4}} -\tfrac{1}{4}e^{-4x} \times 10\,dx\right) \\
&= [-5x^2e^{-4x}]_0^{\frac{1}{4}} + [-\tfrac{5}{2}xe^{-4x}]_0^{\frac{1}{4}} + \int_0^{\frac{1}{4}} \tfrac{5}{2}e^{-4x}\,dx \\
&= [-5x^2e^{-4x}]_0^{\frac{1}{4}} + [-\tfrac{5}{2}xe^{-4x}]_0^{\frac{1}{4}} + [-\tfrac{5}{8}e^{-4x}]_0^{\frac{1}{4}} \\
&= [-5x^2e^{-4x} - \tfrac{5}{2}xe^{-4x} - \tfrac{5}{8}e^{-4x}]_0^{\frac{1}{4}} \\
&= \left(-5(\tfrac{1}{4})^2e^{-4\times\frac{1}{4}} - \tfrac{5}{2}(\tfrac{1}{4})e^{-4\times\frac{1}{4}} - \tfrac{5}{8}e^{-4\times\frac{1}{4}}\right) \\
&\quad - \left(-5 \times 0^2 \times e^{-4\times 0} - \tfrac{5}{2} \times 0 \times e^{-4\times 0} - \tfrac{5}{8}e^{-4\times 0}\right) \\
&= \left(-\tfrac{5}{16}e^{-1} - \tfrac{5}{8}e^{-1} - \tfrac{5}{8}e^{-1}\right) - \left(0 - 0 - \tfrac{5}{8}e^{0}\right) \\
&= \left(-\tfrac{25}{16}e^{-1}\right) - \left(-\tfrac{5}{8}\right) = \tfrac{5}{8} - \tfrac{25}{16}e^{-1}
\end{aligned}$$

26)

$$\int_0^1 xe^x\,dx$$

$$\begin{aligned} u &= x & \frac{dv}{dx} &= e^x \\ \frac{du}{dx} &= 1 & v &= e^x \end{aligned}$$

$$\begin{aligned}
\int_0^1 xe^x\,dx &= [x \times e^x]_0^1 - \int_0^1 e^x \times 1\,dx \\
&= [xe^x]_0^1 - \int_0^1 e^x\,dx \\
&= [xe^x]_0^1 - [e^x]_0^1 \\
&= [xe^x - e^x]_0^1 \\
&= (1 \times e^1 - e^1) - (0 \times e^0 - e^0) \\
&= (e - e) - (0 - 1) = 1
\end{aligned}$$

$$\int_0^1 (2e - x^2e^x)\,dx$$

$$\begin{aligned} u &= x^2 & \frac{dv}{dx} &= e^x \\ \frac{du}{dx} &= 2x & v &= e^x \end{aligned}$$

$$\begin{aligned}
\int_0^1 (2e - x^2e^x)\,dx &= [2ex]_0^1 - \left([x^2 \times e^x]_0^1 - \int_0^1 e^x \times 2x\,dx\right) \\
&= [2ex]_0^1 - \left([x^2e^x]_0^1 - \int_0^1 2xe^x\,dx\right)
\end{aligned}$$

$$= [2ex]_0^1 - [x^2e^x]_0^1 + \int_0^1 2xe^x\,dx$$

$$\begin{array}{ll} u = 2x & \dfrac{dv}{dx} = e^x \\ \dfrac{du}{dx} = 2 & v = e^x \end{array}$$

$$\begin{aligned}
\int_0^1 (2e - x^2e^x)\,dx &= [2ex]_0^1 - [x^2e^x]_0^1 + \int_0^1 2xe^x\,dx \\
&= [2ex]_0^1 - [x^2e^x]_0^1 + \left([2xe^x]_0^1 - \int_0^1 2e^x\,dx\right) \\
&= [2ex]_0^1 - [x^2e^x]_0^1 + [2xe^x]_0^1 - \int_0^1 2e^x\,dx \\
&= [2ex]_0^1 - [x^2e^x]_0^1 + [2xe^x]_0^1 - [2e^x]_0^1 \\
&= [2ex - x^2e^x + 2xe^x - 2e^x]_0^1 \\
&= (2e \times 1 - 1^2 \times e^1 + 2 \times 1 \times e^1 - 2e^1) \\
&\quad - (2e \times 0 - 0^2 \times e^0 + 2 \times 0 \times e^0 - 2e^0) \\
&= (2e - e + 2e - 2e) - (0 - 0 + 0 - 2) = e + 2
\end{aligned}$$

$$\therefore \text{Area} = (e + 2) - 1 = e + 1$$

27)

$$\int_1^e (\ln x)^3\,dx$$

(see answers from 22)

$$\begin{array}{ll} u = \ln x & \dfrac{dv}{dx} = (\ln x)^2 \\ \dfrac{du}{dx} = \dfrac{1}{x} & v = x(\ln x)^2 - 2x\ln x + 2x \end{array}$$

$$\begin{aligned}
\int_1^e (\ln x)^3\,dx &= [\ln x \times (x(\ln x)^2 - 2x\ln x + 2x)]_1^e \\
&\quad - \int_1^e (x(\ln x)^2 - 2x\ln x + 2x) \times \tfrac{1}{x}\,dx \\
&= [x(\ln x)^3 - 2x(\ln x)^2 + 2x\ln x]_1^e - \int_1^e ((\ln x)^2 - 2\ln x + 2)\,dx \\
&= [x(\ln x)^3 - 2x(\ln x)^2 + 2x\ln x]_1^e \\
&\quad - [(x(\ln x)^2 - 2x\ln x + 2x) - 2(x\ln x - x) + 2x]_1^e \\
&= [x(\ln x)^3 - 2x(\ln x)^2 + 2x\ln x - x(\ln x)^2 \\
&\qquad + 2x\ln x - 2x + 2(x\ln x - x) - 2x]_1^e \\
&= [x(\ln x)^3 - 3x(\ln x)^2 + 6x\ln x - 6x]_1^e \\
&= (e \times (\ln e)^3 - 3e \times (\ln e)^2 + 6e \times \ln e - 6e) \\
&\quad - (1 \times (\ln 1)^3 - 3 \times 1 \times (\ln 1)^2 + 6 \times 1 \times \ln 1 - 6 \times 1) \\
&= (e - 3e + 6e - 6e) - (0 - 0 + 0 - 6) \\
&= (-2e) - (-6) = 6 - 2e
\end{aligned}$$

28a)

$$I = \int e^x \sin x \, dx$$

$$\begin{aligned} u &= \sin x & \frac{dv}{dx} &= e^x \\ \frac{du}{dx} &= \cos x & v &= e^x \end{aligned}$$

$$\begin{aligned} I = \int e^x \sin x \, dx &= \sin x \times e^x - \int e^x \times \cos x \, dx \\ &= e^x \sin x - \int e^x \cos x \, dx \end{aligned}$$

$$\begin{aligned} u &= \cos x & \frac{dv}{dx} &= e^x \\ \frac{du}{dx} &= -\sin x & v &= e^x \end{aligned}$$

$$\begin{aligned} I = \int e^x \sin x \, dx &= e^x \sin x - \int e^x \cos x \, dx \\ &= e^x \sin x - \left(\cos x \times e^x - \int e^x \times -\sin x \, dx \right) \\ &= e^x \sin x - \left(e^x \cos x + \int e^x \sin x \, dx \right) \\ &= e^x \sin x - e^x \cos x - \int e^x \sin x \, dx \end{aligned}$$

$$\begin{aligned} \therefore I &= e^x \sin x - e^x \cos x - I \\ \Rightarrow 2I &= e^x \sin x - e^x \cos x \\ \Rightarrow I &= \tfrac{1}{2}(e^x \sin x - e^x \cos x) \\ \therefore \int e^x \sin x \, dx &= \tfrac{1}{2}(e^x \sin x - e^x \cos x) + c \end{aligned}$$

28b)

$$I = \int e^x \cos x \, dx$$

$$\begin{aligned} u &= \cos x & \frac{dv}{dx} &= e^x \\ \frac{du}{dx} &= -\sin x & v &= e^x \end{aligned}$$

$$\begin{aligned} I = \int e^x \cos x \, dx &= \cos x \times e^x - \int e^x \times -\sin x \, dx \\ &= e^x \cos x + \int e^x \sin x \, dx \end{aligned}$$

$$\begin{aligned} u &= \sin x & \frac{dv}{dx} &= e^x \\ \frac{du}{dx} &= \cos x & v &= e^x \end{aligned}$$

$$I = \int e^x \cos x \, dx = e^x \cos x + \int e^x \sin x \, dx$$

$$= e^x \cos x + \left(\sin x \times e^x - \int e^x \cos x \, dx \right)$$
$$= e^x \cos x + \left(e^x \sin x - \int e^x \cos x \, dx \right)$$
$$= e^x \cos x + e^x \sin x - \int e^x \cos x \, dx$$

$$\therefore I = e^x \cos x + e^x \sin x - I$$
$$\Rightarrow 2I = e^x \cos x + e^x \sin x$$
$$\Rightarrow I = \tfrac{1}{2}(e^x \cos x + e^x \sin x)$$
$$\therefore \int e^x \cos x \, dx = \tfrac{1}{2}(e^x \cos x + e^x \sin x) + c$$

29)

$$I = \int_0^{\frac{\pi}{2}} e^{5x} \sin 2x \, dx$$

$$u = \sin 2x \qquad \frac{dv}{dx} = e^{5x}$$
$$\frac{du}{dx} = 2\cos 2x \quad v = \frac{1}{5}e^{5x}$$

$$I = \int_0^{\frac{\pi}{2}} e^{5x} \sin 2x \, dx = [\sin 2x \times \tfrac{1}{5}e^{5x}]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \tfrac{1}{5}e^{5x} \times 2\cos 2x \, dx$$
$$= [\tfrac{1}{5}e^{5x} \sin 2x]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \tfrac{2}{5}e^{5x} \cos 2x \, dx$$

$$u = \cos 2x \qquad \frac{dv}{dx} = \frac{2}{5}e^{5x}$$
$$\frac{du}{dx} = -2\sin 2x \quad v = \frac{2}{25}e^{5x}$$

$$I = \int_0^{\frac{\pi}{2}} e^{5x} \sin 2x \, dx = [\tfrac{1}{5}e^{5x} \sin 2x]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \tfrac{2}{5}e^{5x} \cos 2x \, dx$$
$$= [\tfrac{1}{5}e^{5x} \sin 2x]_0^{\frac{\pi}{2}} - \left([\cos 2x \times \tfrac{2}{25}e^{5x}]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \tfrac{2}{25}e^{5x} \times -2\sin 2x \, dx \right)$$
$$= [\tfrac{1}{5}e^{5x} \sin 2x]_0^{\frac{\pi}{2}} - \left([\tfrac{2}{25}e^{5x} \cos 2x]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} \tfrac{4}{25}e^{5x} \sin 2x \, dx \right)$$
$$= [\tfrac{1}{5}e^{5x} \sin 2x]_0^{\frac{\pi}{2}} - [\tfrac{2}{25}e^{5x} \cos 2x]_0^{\frac{\pi}{2}} - \tfrac{4}{25}\int_0^{\frac{\pi}{2}} e^{5x} \sin 2x \, dx$$

$$\therefore I = [\tfrac{1}{5}e^{5x} \sin 2x - \tfrac{2}{25}e^{5x} \cos 2x]_0^{\frac{\pi}{2}} - \tfrac{4}{25}I$$
$$\Rightarrow \tfrac{29}{25}I = [\tfrac{1}{5}e^{5x} \sin 2x - \tfrac{2}{25}e^{5x} \cos 2x]_0^{\frac{\pi}{2}}$$
$$\Rightarrow I = [\tfrac{5}{29}e^{5x} \sin 2x - \tfrac{2}{29}e^{5x} \cos 2x]_0^{\frac{\pi}{2}}$$

$$= \left(\tfrac{5}{29}e^{5\times\frac{\pi}{2}} \sin(2 \times \tfrac{\pi}{2}) - \tfrac{2}{29}e^{5\times\frac{\pi}{2}} \cos(2 \times \tfrac{\pi}{2}) \right)$$
$$- \left(\tfrac{5}{29}e^{5\times 0} \sin(2 \times 0) - \tfrac{2}{29}e^{5\times 0} \cos(2 \times 0) \right)$$
$$= \left(0 + \tfrac{2}{29}e^{\frac{5\pi}{2}} \right) - \left(0 - \tfrac{2}{29} \right) = \tfrac{2}{29}(1 + e^{\frac{5\pi}{2}})$$

30)

Passes through the origin: $0 = \ln(2 \times 0 - k)$

$$\Rightarrow -k = e^0 = 1 \Rightarrow k = -1$$

$$y = \ln(2x+1) \Rightarrow \frac{dy}{dx} = \frac{2}{2x+1}$$

A has coordinates $(1, \ln 3)$

$$\left.\frac{dy}{dx}\right|_{x=1} = \frac{2}{2 \times 1 + 1} = \frac{2}{3}$$

Normal is $y - \ln 3 = -\frac{3}{2}(x-1)$

$$\text{When } y = 0, \quad 0 - \ln 3 = -\tfrac{3}{2}(x-1)$$
$$\Rightarrow -\ln 3 = -\tfrac{3}{2}x + \tfrac{3}{2} \Rightarrow 3x = 2\ln 3 + 3 \Rightarrow x = \tfrac{2}{3}\ln 3 + 1$$
$$\therefore B \text{ has coordinates } \left(\tfrac{2}{3}\ln 3 + 1, 0\right)$$

$$\int_0^1 \ln(2x+1)\,dx$$

$$u = \ln(2x+1) \quad \frac{dv}{dx} = 1$$
$$\frac{du}{dx} = \frac{2}{2x+1} \quad v = x$$

$$\int_0^1 \ln(2x+1)\,dx = [\ln(2x+1) \times x]_0^1 - \int_0^1 x \times \tfrac{2}{2x+1}\,dx$$
$$= [x\ln(2x+1)]_0^1 - \int_0^1 \tfrac{2x}{2x+1}\,dx$$

Focus on $\displaystyle\int_0^1 \frac{2x}{2x+1}\,dx$ and use Integration by Substitution

$$u = 2x+1 \Rightarrow \frac{du}{dx} = 2 \Rightarrow dx = \tfrac{1}{2}du$$
$$u = 2x+1 \Rightarrow 2x = u - 1$$

Change limits:

$$x = 1 \Rightarrow u = 2 \times 1 + 1 = 3$$
$$x = 0 \Rightarrow u = 2 \times 0 + 1 = 1$$

$$\int_0^1 \frac{2x}{2x+1}\,dx = \int_1^3 \frac{u-1}{u} \times \tfrac{1}{2}\,du$$
$$= \int_1^3 \tfrac{1}{2}\left(\tfrac{u}{u} - \tfrac{1}{u}\right)\,du = \int_1^3 \left(\tfrac{1}{2} - \tfrac{1}{2u}\right)\,du$$
$$= [\tfrac{1}{2}u - \tfrac{1}{2}\ln|u|]_1^3$$
$$= \left(\tfrac{1}{2} \times 3 - \tfrac{1}{2}\ln|3|\right) - \left(\tfrac{1}{2} \times 1 - \tfrac{1}{2}\ln|1|\right)$$
$$= \left(\tfrac{3}{2} - \tfrac{1}{2}\ln 3\right) - \left(\tfrac{1}{2} - \tfrac{1}{2}\ln 1\right)$$
$$= \left(\tfrac{3}{2} - \tfrac{1}{2}\ln 3\right) - \left(\tfrac{1}{2} - 0\right) = 1 - \tfrac{1}{2}\ln 3$$

$$\therefore \int_0^1 \ln(2x+1)\,dx = [x\ln(2x+1)]_0^1 - \left(1 - \tfrac{1}{2}\ln 3\right)$$

$$= (1 \times \ln(2 \times 1 + 1)) - (0 \times \ln(2 \times 0 + 1)) - 1 + \tfrac{1}{2} \ln 3$$
$$= (\ln 3) - (0) - 1 + \tfrac{1}{2} \ln 3 = \tfrac{3}{2} \ln 3 - 1$$

Area of Triangle $= \tfrac{1}{2} \times \tfrac{2}{3} \ln 3 \times \ln 3 = \tfrac{1}{3} (\ln 3)^2$

$$\therefore \text{Area} = \left(\tfrac{3}{2} \ln 3 - 1 \right) + \tfrac{1}{3} (\ln 3)^2$$
$$= \tfrac{1}{3} (\ln 3)^2 + \tfrac{3}{2} \ln 3 - 1$$